

Nathan Kong

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Education

Penn State University
B.S. in Computer Engineering

Expected Graduation: May 2027

Work Experience

Penn State Autonomous Vehicle Team – Perception Department

January 2026 – May 2026

- Participating in Autodrive Challenge using YOLO-based architecture with **PyTorch** and **TensorRT**.
- Reduced end to end obstacle detection by **40%** by offloading CPU tasks to GPU.
- Successfully engineered a highly reliable Lane Detection with full integration with ROS2 based system stack; this achievement is a 1st in the program.
- Managed full-cycle ML workflows, including custom dataset curation and loss function tuning to mitigate model regression under high stakes environment.
- Validated perception models and lane detection algorithms by replaying ROS2 bags from live track tests to identify use cases. Utilized internal analysis tools to identify regression and ensure software reliability in high stakes environment.

Hexagon – Software Engineer Intern

May – August 2025

- Engineered a high-scale **Python** framework with asynchronous pipelines to handle complex business logic, optimizing internal data flows and improving retrieval speed via **Azure** APIs.

Independent Research

High-Performance GPU Kernel Optimization

Summer 2025

- Developed and optimized fused attention mechanisms in **Triton**, achieving **8x** throughput gains on **NVIDIA A100s** via persistent kernel execution and warp specialization.
- Optimized memory locality and tiling strategies to minimize global memory bottlenecks, directly improving long-context inference performance for Transformer-based models.
- Conducted deep-dive benchmarks on memory-bound vs. compute-bound tradeoffs, analyzing **SRAM bandwidth** and GPU utilization to ensure hardware-aware model execution.

Technical Projects

Neural Network with Backpropagation

Spring 2026

- Established a baseline MNIST classifier using **PyTorch** to validate network architecture and set benchmarks for prediction accuracy.
- Implemented back propagation without ML framework.
- Optimized training convergence by implementing custom Gradient Descent variations, achieving higher accuracy in significantly fewer iterations than the baseline.
- Rewrite to **C++** with custom memory management and linear algebra to align data with CPU/GPU cache and minimize latency.
- Achieved and measured raw speedups of **13x** through hardware-software co-design, demonstrating that hardware-specific tuning can bypass the inherent limitations of general-purpose CUDA.

Elevator Controller: Python to Vivado HLS

Fall 2025

- SW Solution
 - Formulated a Finite State Machine and turned it into pseudocode
 - Engineered Python based simulation and testbench to validate FSM against defined use cases.
 - Implemented **C++** firmware ensuring execution of FSM logic while interfacing with physical IMU sensors.
- HW Solution
 - Prototyped a microcontroller-based system by interfacing analog sensors with custom logic schematics.
 - Validated physical component responses against defined use cases to ensure hardware-software consistency.
 - Cross-verified real-time sensor data against **Python** testbench models to ensure FSM state accuracy.

Skills

- **Languages:** C++, Python, Triton, RTL, CUDA, Bash, CMAKE
- **Systems/Tools:** ROS2, TensorRT, Linux, Docker, Vivado SW tools Git, Azure